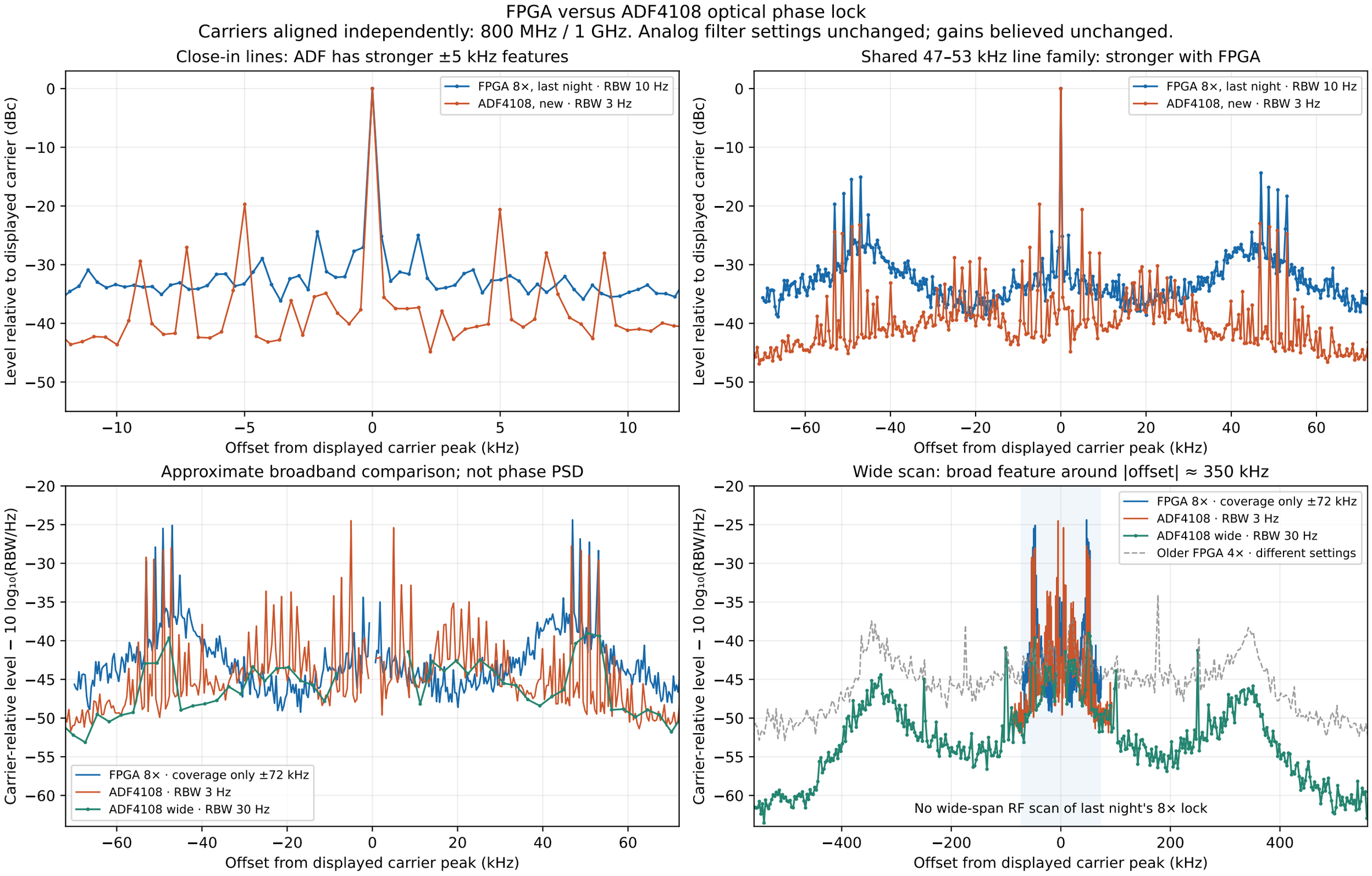

FPGA versus ADF4108 optical phase lock
Carriers aligned independently: 800 MHz / 1 GHz. Analog filter settings unchanged; gains believed unchanged.
Close-in lines: ADF has stronger ±5 kHz features
Shared 47–53 kHz line family: stronger with FPGA
Approximate broadband comparison; not phase PSD
Wide scan: broad feature around |offset| ≈ 350 kHz
FPGA 8×, last night · RBW 10 Hz
ADF4108, new · RBW 3 Hz
FPGA 8× · coverage only ±72 kHz
ADF4108 · RBW 3 Hz
ADF4108 wide · RBW 30 Hz
Older FPGA 4× · different settings
No wide-span RF scan of last night's 8× lock
Level relative to displayed carrier (dBc)
Carrier-relative level − 10 log₁₀(RBW/Hz)
Offset from displayed carrier peak (kHz)
Positive-peak detection, one sweep, incomplete-sweep flags. Native line heights and RBW-normalized backgrounds have different meanings.